

# 1. Briefly describe any four normality tests commonly used in hypothesis testing

## Types of normality tests

### 1. Shapiro-Wilk Test for normality

- Many statistical tests assume the data is normally distributed.
- A statistical test designed to assess whether a sample comes from a normally distributed population.
- Suitable for small sample sizes ( $n < 50$ ), but can be applied to larger datasets as well (but not advised to do so).
- Null vs Alternative Hypothesis:
  - NULL Hypothesis ( $H_0$ ): The data is normally distributed
  - Alternative Hypothesis ( $H_1$ ): The data is not normally distributed
  - We reject the null hypothesis if the P value is less than Alpha
  - Typically set alpha to 0.05

### 2. Kolmogorov-Smirnov Test

- A **non-parametric test** that compares the **empirical distribution function (EDF)** of the sample to a **fully specified theoretical distribution**.
- It calculates the **maximum absolute difference** between the EDF and the theoretical cumulative distribution function (CDF).
- Used to test whether a sample comes from a specified continuous distribution.
- In this case it compares EDF to normal distribution for normality tests.

### 3. Anderson-Darling test

- It is a refinement of the Kolmogorov-Smirnov test, but it gives **more weight to the tails** of the distribution.
- This makes it **more sensitive to deviations** from normality that occur in the tails (e.g., heavy tails, skewed distributions).

### 4. D'Agostino $K^2$ test

- It works by examining **two key features of a distribution**:
- **Skewness** – measures asymmetry (left- or right-skewed).
- **Kurtosis** – measures how heavy or light the tails are compared to a normal distribution.
- The test transforms the sample skewness and kurtosis into **z-scores**, then combines them to form the test statistic  **$K^2$** .

- If the skewness or kurtosis deviates significantly from what is expected under normality, the test rejects the null hypothesis of normality.

## 2. Match each normality test to the appropriate sample size range (n)

| Sample Size Range           | Appropriate Test(s)                    |
|-----------------------------|----------------------------------------|
| (a) $n < 50$                | Shapiro–Wilk test                      |
| (b) $50 < n < 100$          | Anderson–Darling test, D’Agostino test |
| (c) $n > 5000$ (very large) | D’Agostino $K^2$ test                  |

## 3. Explain under what circumstances normality tests should be avoided

Normality tests should be avoided in the following situations:

(a) When the sample size is very large

- For large  $n$ , even trivial deviations from normality become statistically significant.
- The test may reject normality even when the distribution is practically close enough for parametric tests.

(b) When the sample size is very small

- Test power is too weak to detect meaningful deviations.
- Visual tools (Q–Q plots, histograms) are often more informative.

(c) When the downstream analysis is robust to non-normality

- Many tests (e.g., t-tests with reasonably large  $n$ , ANOVA under equal variances) are robust to mild deviations.
- In such cases, testing for normality may be unnecessary and misleading.

(d) When data are known not to be from a continuous distribution

- Normality tests assume continuous data; discrete or categorical data violate assumptions.

(e) When normality is imposed unnecessarily

- Some analyses (e.g., nonparametric tests, GLMs) do not require normality of the raw data.